

MOSFET – Complementary, POWERTRENCH®

60 V

FDS4559

General Description

This complementary MOSFET device is produced using onsemi's advanced PowerTrench process that has been especially tailored to minimize the on-state resistance and yet maintain low gate charge for superior switching performance.

Features

- Q1: N-Channel
 - 4.5 A, 60 V
 - $R_{DS(on)} = 55\text{ m}\Omega @ V_{GS} = 10\text{ V}$
 - $R_{DS(on)} = 75\text{ m}\Omega @ V_{GS} = 4.5\text{ V}$
- Q2: P-Channel
 - 3.5 A, -60 V
 - $R_{DS(on)} = 105\text{ m}\Omega @ V_{GS} = -10\text{ V}$
 - $R_{DS(on)} = 135\text{ m}\Omega @ V_{GS} = -4.5\text{ V}$

Applications

- DC/DC converter
- Power management
- LCD backlight inverter
- This is a Pb-Free and Halide Free Device

ABSOLUTE MAXIMUM RATINGS ($T_A = 25^\circ\text{C}$ unless otherwise noted)

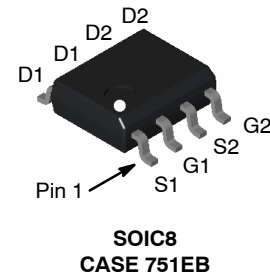
Symbol	Parameter	Q1	Q2	Unit
V_{DSS}	Drain-Source Voltage	60	-60	V
V_{GSS}	Gate-Source Voltage	± 20	± 20	V
I_D	Drain Current	Continuous (Note 1a)	4.5	A
		Pulsed	20	
P_D	Power Dissipation for Dual Operation		2	W
	Power Dissipation for Single Operation	(Note 1a)	1.6	
		(Note 1b)	1.2	
		(Note 1c)	1	
T_J, T_{STG}	Operating and Storage Junction Temperature Range	-55 to +175		$^\circ\text{C}$

Stresses exceeding those listed in the Maximum Ratings table may damage the device. If any of these limits are exceeded, device functionality should not be assumed, damage may occur and reliability may be affected.

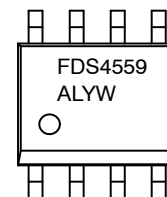
THERMAL CHARACTERISTICS

Symbol	Parameter	Value	Unit
$R_{\theta JA}$	Thermal Resistance, Junction-to-Ambient (Note 1a)	78	$^\circ\text{C/W}$
$R_{\theta JC}$	Thermal Resistance, Junction-to-Case (Note 1)	40	$^\circ\text{C/W}$

V_{DSS}	$R_{DS(on)}$ Max	I_D Max
N-Channel 60 V	55 m Ω @ 10 V	4.5 A
	75 m Ω @ 4.5 V	
P-Channel -60 V	105 m Ω @ -10 V	-3.5 A
	135 m Ω @ -4.5 V	

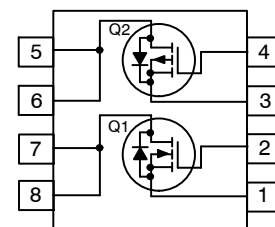


MARKING DIAGRAM



FDS4559 = Specific Device Code
 A = Assembly Site
 L = Wafer Lot Number
 YW = Assembly Start Week

N-Channel / P-Channel



ORDERING INFORMATION

Device	Package	Shipping†
FDS4559	SOIC8 (Pb-Free, Halide Free)	2500 / Tape & Reel

†For information on tape and reel specifications, including part orientation and tape sizes, please refer to our Tape and Reel Packaging Specifications Brochure, BRD8011/D.

ELECTRICAL CHARACTERISTICS ($T_A = 25^\circ\text{C}$ unless otherwise noted)

Symbol	Parameter	Test Condition	Typ	Min	Typ	Max	Unit
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DRAIN-SOURCE AVALANCHE RATINGS (Note 1)

W_{DSS}	Single Pulse Drain-Source Avalanche Energy	$V_{DD} = 30\text{ V}$, $I_D = 25\text{ A}$	Q1	–	–	90	V
I_{AR}	Maximum Drain-Source Avalanche Current		Q1	–	–	4.5	V

OFF CHARACTERISTICS

BV_{DSS}	Drain-Source Breakdown Voltage	$V_{GS} = 0\text{ V}$, $I_D = 250\text{ }\mu\text{A}$ $V_{GS} = 0\text{ V}$, $I_D = -250\text{ }\mu\text{A}$	Q1 Q2	60 –60	– –	– –	V
$\frac{\Delta BV_{DSS}}{\Delta T_J}$	Breakdown Voltage Temperature Coefficient	$I_D = 250\text{ }\mu\text{A}$, Referenced to 25°C $I_D = -250\text{ }\mu\text{A}$, Referenced to 25°C	Q1 Q2	– –	58 –49	– –	$\text{mV}/^\circ\text{C}$
I_{DSS}	Zero Gate Voltage Drain Current	$V_{DS} = 48\text{ V}$, $V_{GS} = 0\text{ V}$ $V_{DS} = -48\text{ V}$, $V_{GS} = 0\text{ V}$	Q1 Q2	– –	– –	1 –1	μA
I_{GSS}	Gate-Body Leakage	$V_{GS} = \pm 20\text{ V}$, $V_{DS} = 0\text{ V}$ $V_{GS} = \pm 20\text{ V}$, $V_{DS} = 0\text{ V}$	Q1 Q2	– –	– –	± 100 ± 100	nA

ON CHARACTERISTICS (Note 2)

$V_{GS(th)}$	Gate Threshold Voltage	$V_{DS} = V_{GS}$, $I_D = 250\text{ }\mu\text{A}$ $V_{DS} = V_{GS}$, $I_D = -250\text{ }\mu\text{A}$	Q1 Q2	1 –1	2.2 –1.6	3 –3	V
$\frac{\Delta V_{GS(th)}}{\Delta T_J}$	Gate Threshold Voltage Temperature Coefficient	$I_D = 250\text{ }\mu\text{A}$, Referenced to 25°C $I_D = -250\text{ }\mu\text{A}$, Referenced to 25°C	Q1 Q2	– –	–5.5 4	– –	$\text{mV}/^\circ\text{C}$
$R_{DS(on)}$	Static Drain-Source On-Resistance	$V_{GS} = 10\text{ V}$, $I_D = 4.5\text{ A}$ $V_{GS} = 10\text{ V}$, $I_D = 4.5\text{ A}$, $T_j = 125^\circ\text{C}$ $V_{GS} = 4.5\text{ V}$, $I_D = 4\text{ A}$	Q1	– – –	42 72 55	55 94 75	$\text{m}\Omega$
		$V_{GS} = -10\text{ V}$, $I_D = -3.5\text{ A}$ $V_{GS} = -10\text{ V}$, $I_D = -3.5\text{ A}$, $T_j = 125^\circ\text{C}$ $V_{GS} = -4.5\text{ V}$, $I_D = -3.1\text{ A}$	Q2	– – –	82 130 105	105 190 135	$\text{m}\Omega$
$I_{D(on)}$	On-State Drain Current	$V_{GS} = 10\text{ V}$, $V_{DS} = 5\text{ V}$ $V_{GS} = -10\text{ V}$, $V_{DS} = -5\text{ V}$	Q1 Q2	20 –20	– –	– –	A
g_{FS}	Forward Transconductance	$V_{DS} = 10\text{ V}$, $I_D = 4.5\text{ A}$ $V_{DS} = -5\text{ V}$, $I_D = -3.5\text{ A}$	Q1 Q2	– –	14 9	– –	S

DYNAMIC CHARACTERISTICS

C_{iss}	Input Capacitance	Q1 $V_{DS} = 25\text{ V}$, $V_{GS} = 0\text{ V}$, $f = 1.0\text{ Mhz}$ Q2 $V_{DS} = -30\text{ V}$, $V_{GS} = 0\text{ V}$, $f = 1.0\text{ MHz}$	Q1 Q2	– –	650 759	– –	pF
C_{oss}	Output Capacitance		Q1 Q2	– –	80 90	– –	pF
C_{rss}	Reverse Transfer Capacitance		Q1 Q2	– –	35 39	– –	pF

SWITCHING CHARACTERISTICS (Note 2)

$t_{d(on)}$	Turn-On Delay Time	Q1 $V_{DD} = 30\text{ V}$, $I_D = 1\text{ A}$, $V_{GS} = 10\text{ V}$, $R_{GEN} = 6\text{ }\Omega$ Q2 $V_{DD} = -30\text{ V}$, $I_D = -1\text{ A}$, $V_{GS} = -10\text{ V}$, $R_{GEN} = 6\text{ }\Omega$	Q1 Q2	– –	11 7	20 14	ns
t_r	Turn-On Rise Time		Q1 Q2	– –	8 10	18 20	ns
$t_{d(off)}$	Turn-Off Delay Time		Q1 Q2	– –	19 19	35 34	ns
t_f	Turn-Off Fall Time		Q1 Q2	– –	6 12	15 22	ns
Q_g	Total Gate Charge	Q1 $V_{DD} = 30\text{ V}$, $I_D = 4.5\text{ A}$, $V_{GS} = 10\text{ V}$ Q2 $V_{DD} = -30\text{ V}$, $I_D = -3.5\text{ A}$, $V_{GS} = -10\text{ V}$	Q1 Q2	– –	12.5 15	18 21	nC
Q_{gs}	Gate-Source Charge		Q1 Q2	– –	2.4 2.5	– –	nC
Q_{gd}	Gate-Drain Charge		Q1 Q2	– –	2.6 3.0	– –	nC

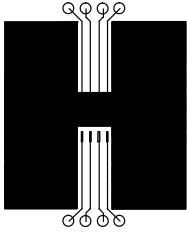
ELECTRICAL CHARACTERISTICS ($T_A = 25^\circ\text{C}$ unless otherwise noted)

DRAIN-SOURCE DIODE CHARACTERISTICS AND MAXIMUM RATINGS

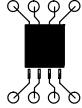
I_S	Maximum Continuous Drain-Source Diode Forward Current		Q1 Q2	- -	- -	1.3 -1.3	A
V_{SD}	Drain-Source Diode Forward Voltage	$V_{GS} = 0\text{ V}, I_S = 1.3\text{ A}$ (Note 2) $V_{GS} = 0\text{ V}, I_S = -1.3\text{ A}$ (Note 2)	Q1 Q2	- -	0.8 -0.8	1.2 -1.2	V

Product parametric performance is indicated in the Electrical Characteristics for the listed test conditions, unless otherwise noted. Product performance may not be indicated by the Electrical Characteristics if operated under different conditions.

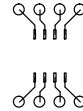
1. $R_{\theta JA}$ is the sum of the junction-to-case and case-to-ambient thermal resistance where the case thermal reference is defined as the solder mounting surface of the drain pins. $R_{\theta JC}$ is guaranteed by design while $R_{\theta CA}$ is determined by the user's board design.



a) 78°C/W when mounted on a 0.5 in^2 pad of 2 oz copper



b) 125°C/W when mounted on a $.02\text{ in}^2$ pad of 2 oz copper



c) 135°C/W when mounted on a minimum pad

Scale 1:1 on letter size paper

2. Pulse Test: Pulse Width $< 300\text{ }\mu\text{s}$, Duty Cycle $< 2.0\%$

TYPICAL CHARACTERISTICS (Q2 P-CHANNEL)

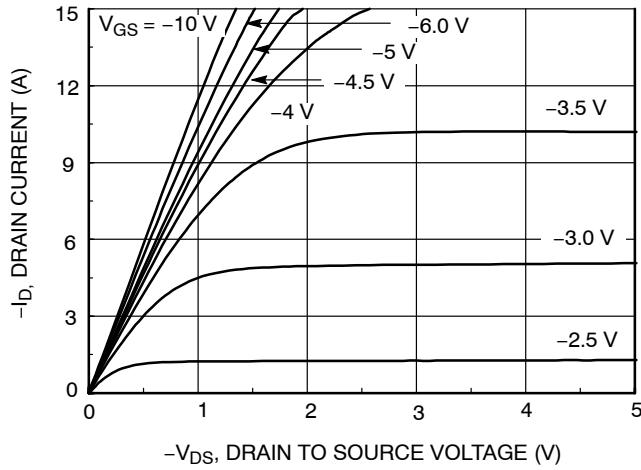


Figure 1. On-Region Characteristics

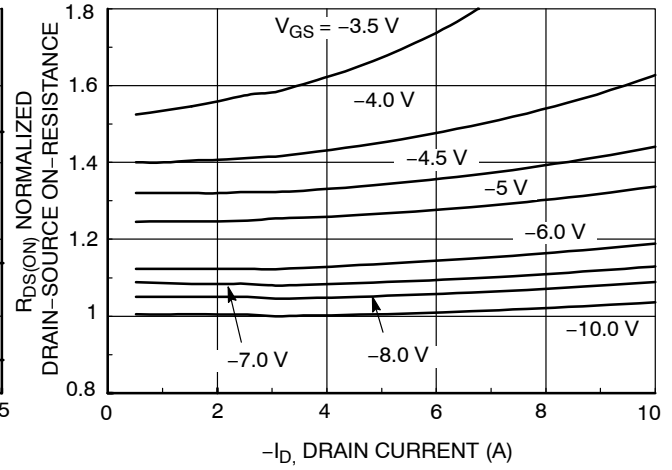


Figure 2. On-Resistance Variation with Drain Current and Gate Voltage

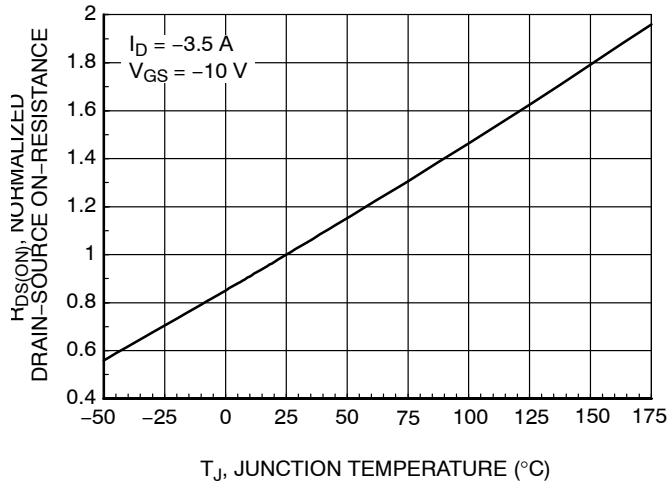


Figure 3. On-Resistance Variation with Temperature

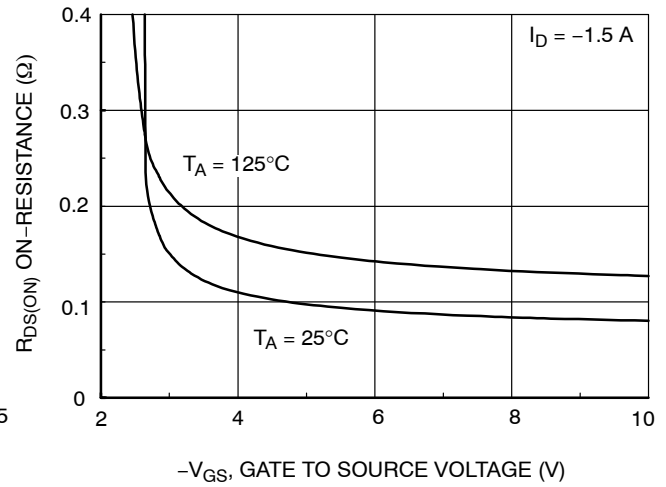


Figure 4. On-Resistance Variation with Gate-to-Source Voltage

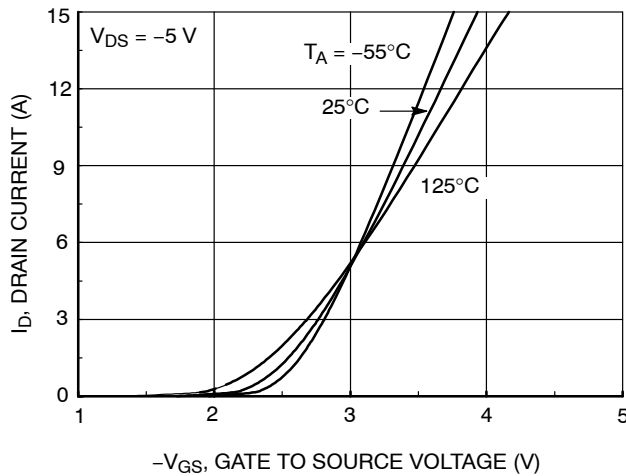


Figure 5. Transfer Characteristics

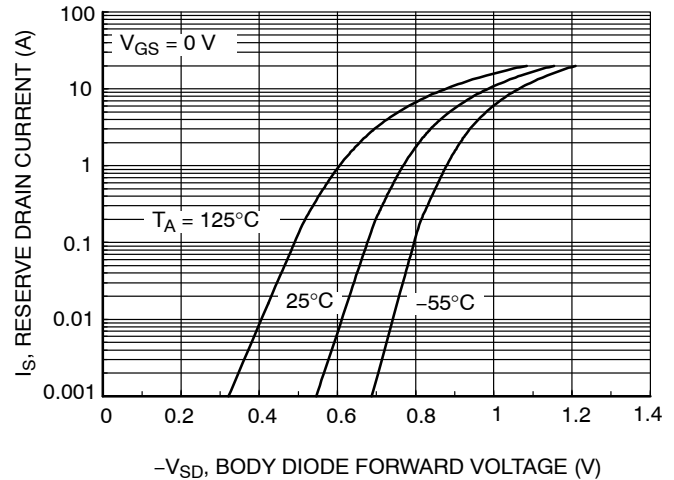


Figure 6. Body Diode Forward Voltage Variation with Source Current and Temperature

TYPICAL CHARACTERISTICS (Q2 P-CHANNEL) (continued)

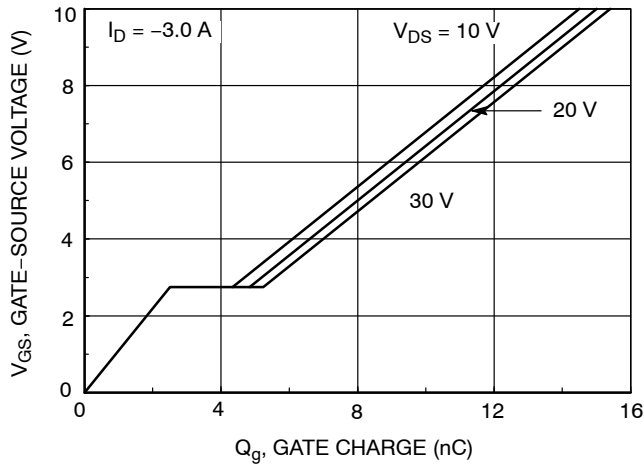


Figure 7. Gate Charge Characteristics

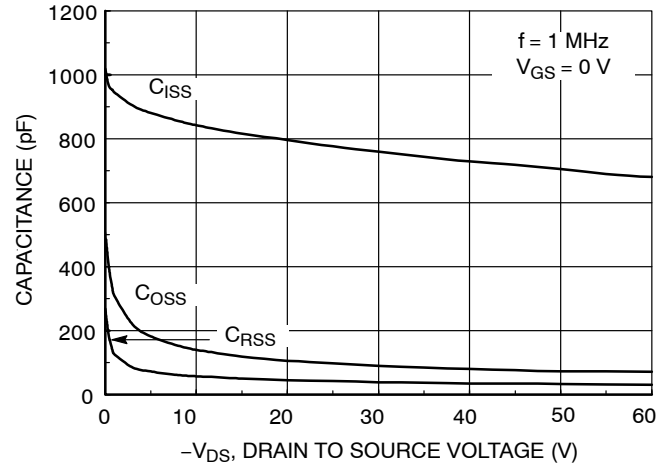


Figure 8. Capacitance Characteristics

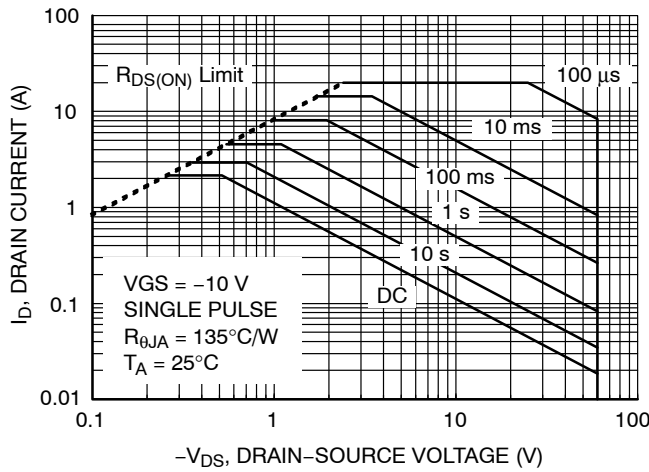


Figure 9. Maximum Safe Operating Area

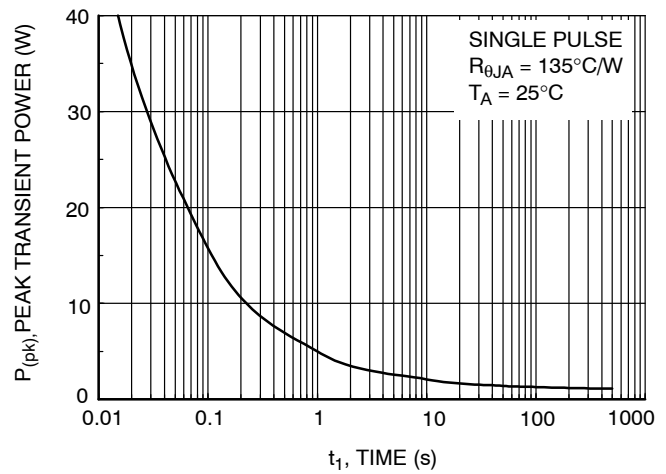


Figure 10. Single Pulse Maximum Power Dissipation

TYPICAL CHARACTERISTICS (Q1 N-CHANNEL)

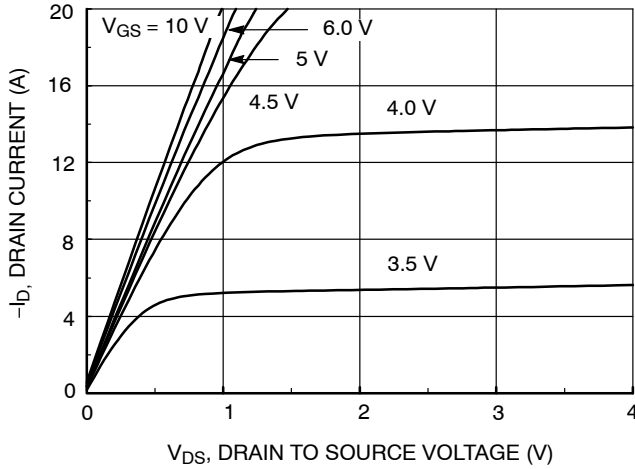


Figure 11. On-Region Characteristics

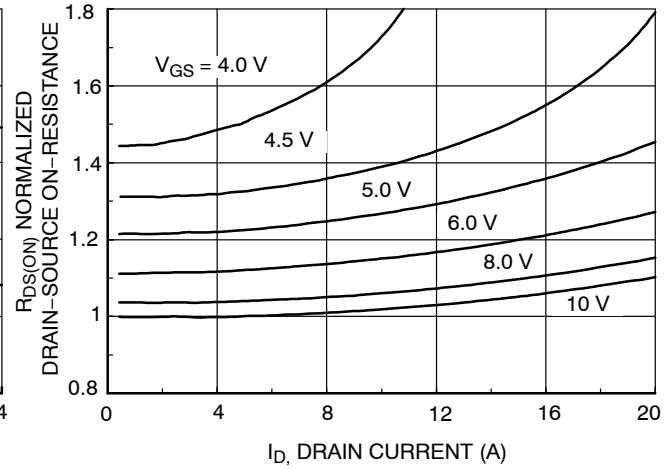


Figure 12. On-Resistance Variation with Drain Current and Gate Voltage

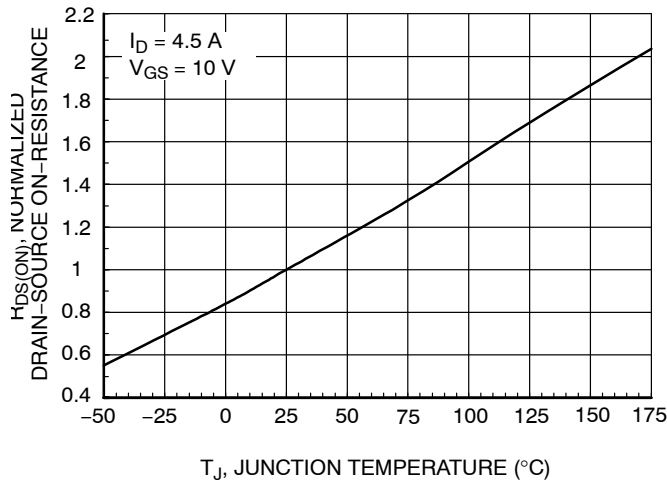


Figure 13. On-Resistance Variation with Temperature

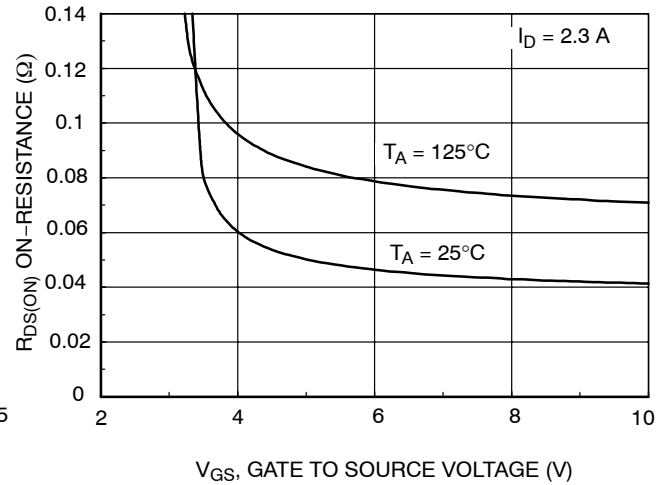


Figure 14. On-Resistance Variation with Gate-to-Source Voltage

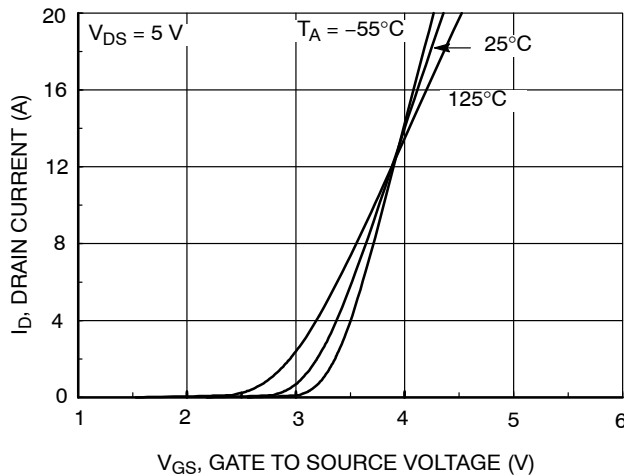


Figure 15. Transfer Characteristics

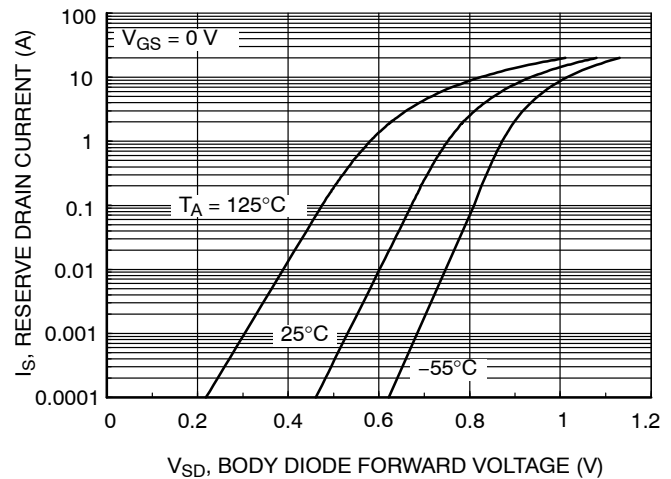


Figure 16. Body Diode Forward Voltage Variation with Source Current and Temperature

TYPICAL CHARACTERISTICS (Q1 N-CHANNEL) (continued)

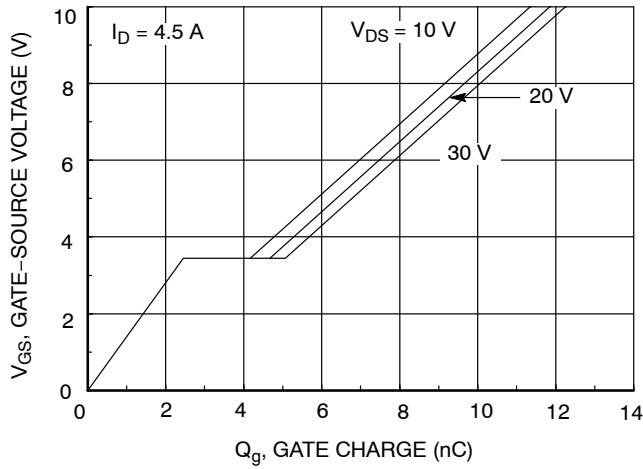


Figure 17. Gate Charge Characteristics

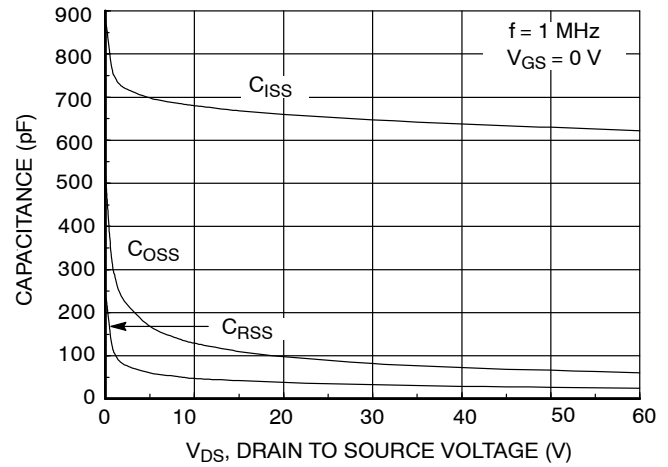


Figure 18. Capacitance Characteristics

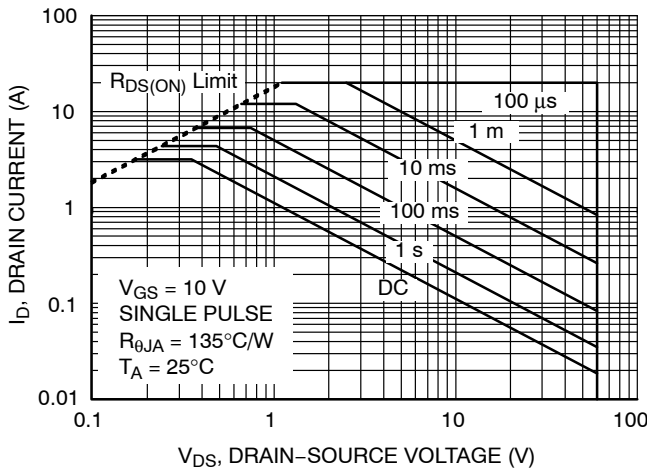


Figure 19. Maximum Safe Operating Area

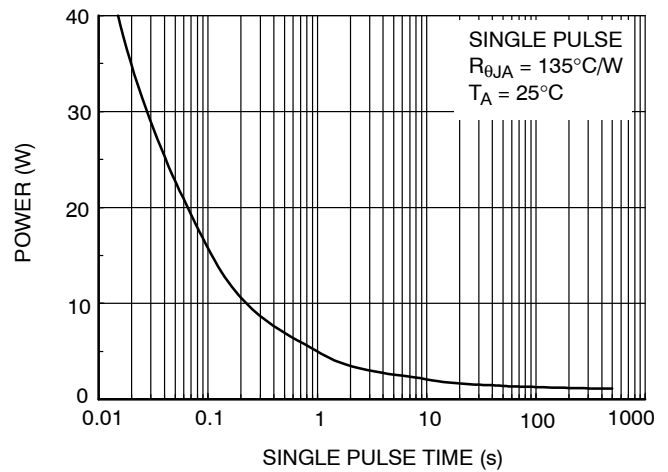


Figure 20. Single Pulse Maximum Power Dissipation

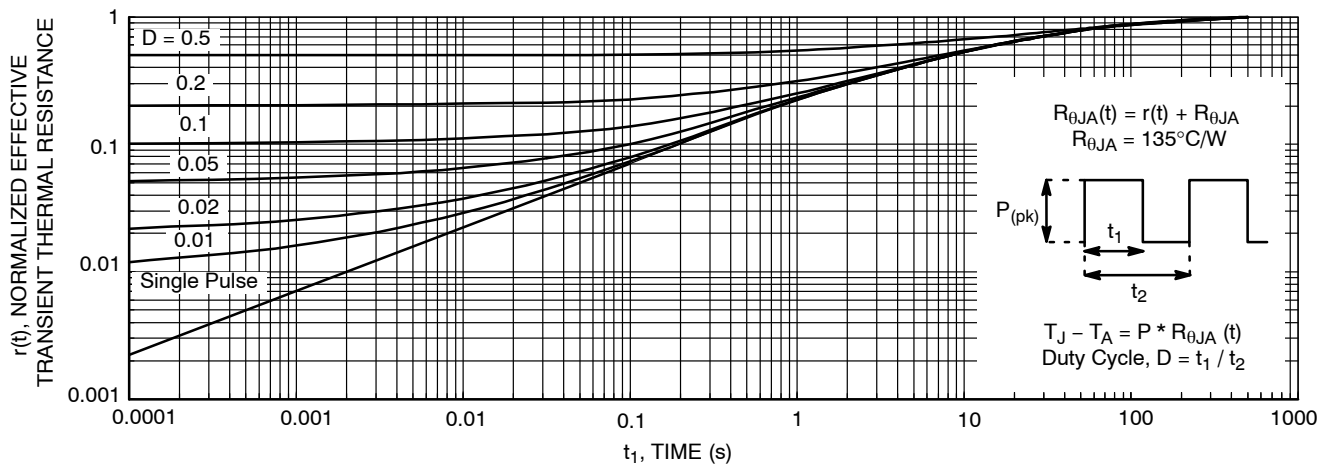


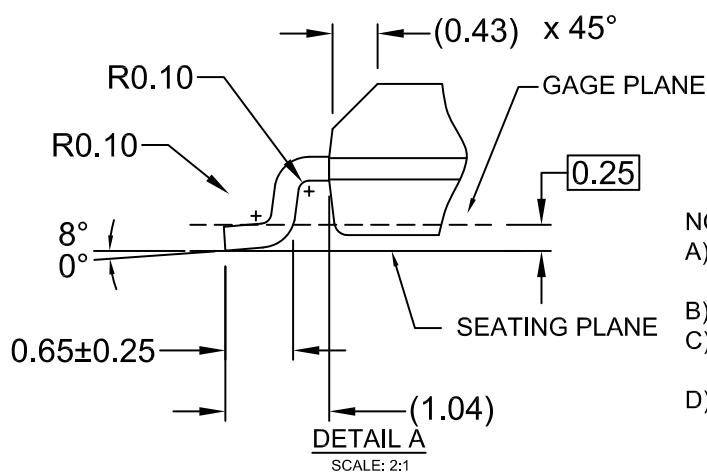
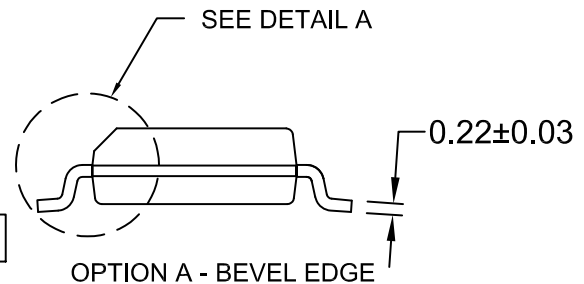
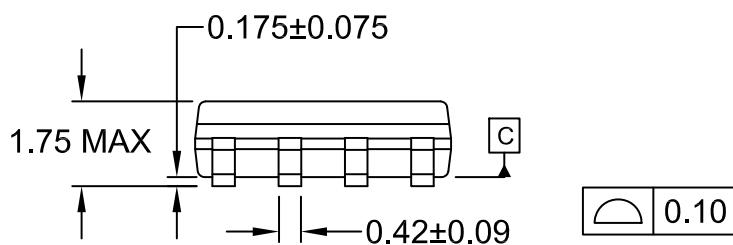
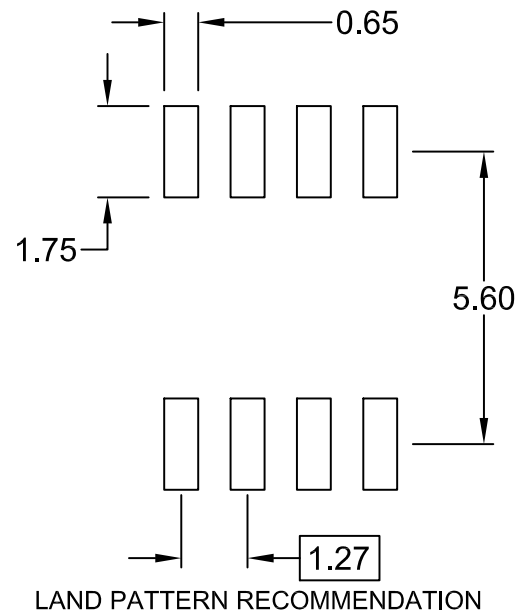
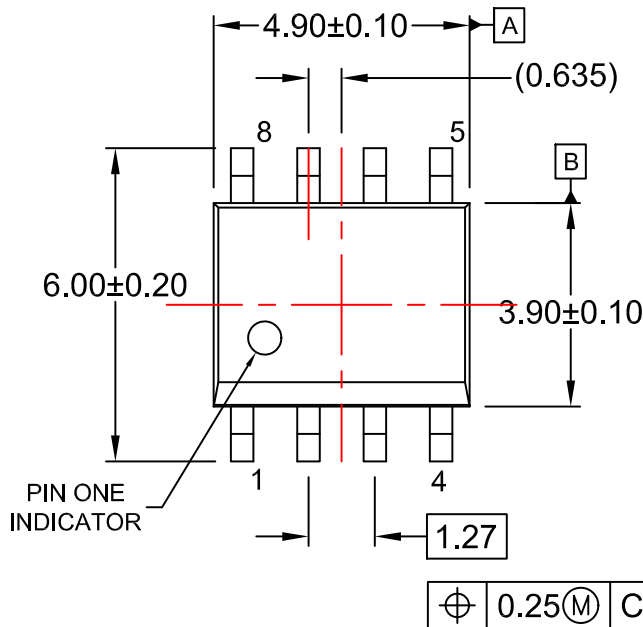
Figure 21. Transient Thermal Response Curve

Thermal characterization performed using the conditions described in Note 1c.
Transient thermal response will change depending on the circuit board design.

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SOIC8
CASE 751EB
ISSUE A

DATE 24 AUG 2017



NOTES:

- A) THIS PACKAGE CONFORMS TO JEDEC MS-012, VARIATION AA.
- B) ALL DIMENSIONS ARE IN MILLIMETERS.
- C) DIMENSIONS DO NOT INCLUDE MOLD FLASH OR BURRS.
- D) LANDPATTERN STANDARD: SOIC127P600X175-8M

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